

M.KUMARASAMY COLLEGE OF ENGINEERING , KARUR.

**Department of Electronics and Communication
Engineering**

EGB1221-DATABASE MANAGEMENT SYSTEM

IV Semester (2023-2027 Batch)

FINAL REVIEW PRESENTATION

Register Number :927623BEC101,927623BEC102

Name :KANISHKA.D , KANISHKA.S

Date :03.06.2025

TITLE OF THE PROJECT

LIBRARY MANAGEMENT SYSTEM

Abstract

A Library Management System (LMS) is a robust and essential application of Database Management Systems (DBMS) that enables efficient and systematic control over library resources. In a traditional library, the manual tracking of books, members, and transactions often results in errors, redundancy, and time-consuming operations. To overcome these limitations, the LMS integrates a well-structured DBMS to store, manage, and retrieve library data quickly and reliably. Each entity is represented by a table, and relationships are established among them using foreign key constraints. For example, a 'Books' table might include attributes such as Book_ID, Title, Author, Publisher, and Availability_Status.

Literature Review

Year	Authors	Title of the Paper	Algorithm/ Methodologies/ Techniques	Findings
2023	Jing Wangetal	Study on Library Management System Based on Data Mining and Clustering Algorithm	Big Data Fusion, Clustering Algorithms	May not generalize well across diverse datasets or libraries with different user behaviors.
2020	Anonymous (Discover Applied Sciences)	Application of Machine Learning Algorithm in Electronic Book Database Management system	Machine Learning(Linear Regression Model)	Possible difficulty in handling multi_x0002_language or multimedia-rich content in e_x0002_books.

Problem(s) Identified

A Library Management System (LMS) is required to efficiently manage the operations of a library. The system must facilitate the management of books, members, transactions (issue and return of books), and other related activities. The system should maintain an organized database and allow authorized personnel (librarians, administrators) to interact with it seamlessly

Proposed Solution

To address the challenges of manual library operations, we propose a digital Library Management System (LMS) that offers an intuitive, user-friendly interface for librarians and users. The system will be built with secure login-based access for both administrators (librarians) and general users.

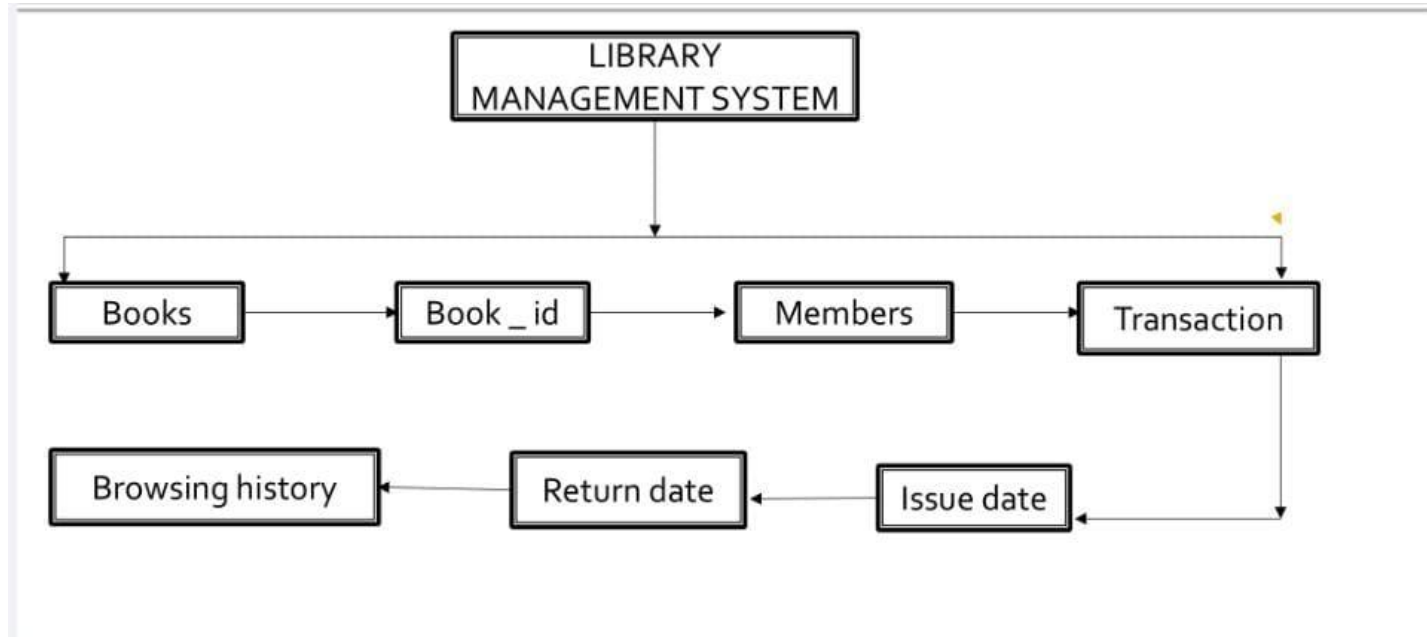
The LMS will include the following key modules:

1. Book Management – Allows librarians to add, edit, delete, and categorize books in the catalog.
2. User Management – Enables registration and maintenance of user records, including borrowing history.
3. Search & Reservation – Provides users with advanced search tools to check book availability and reserve books online.
4. Issuance & Returns – Automates the issuing and returning process, tracks due dates, and calculates fines for overdue items

Objectives

The Library Management System is designed to provide a user-friendly interface for librarians to efficiently manage books, user records, and all related transactions. Through this system, librarians can add, update, or remove books, register new users, and oversee the issuing and returning of materials. Users are given access to search the library catalog, check book availability, and reserve or borrow items directly through the system. To ensure smooth operation and avoid common issues, the system incorporates rules and alerts to prevent double issuance of books and to notify users and staff about overdue returns.

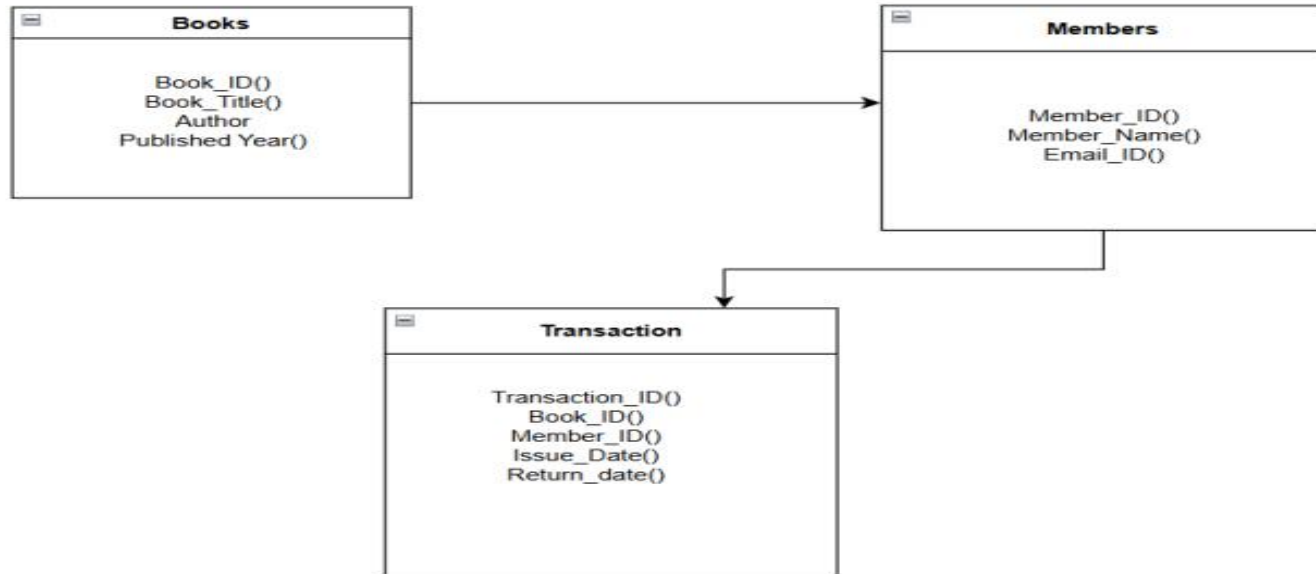
Architecture Diagram



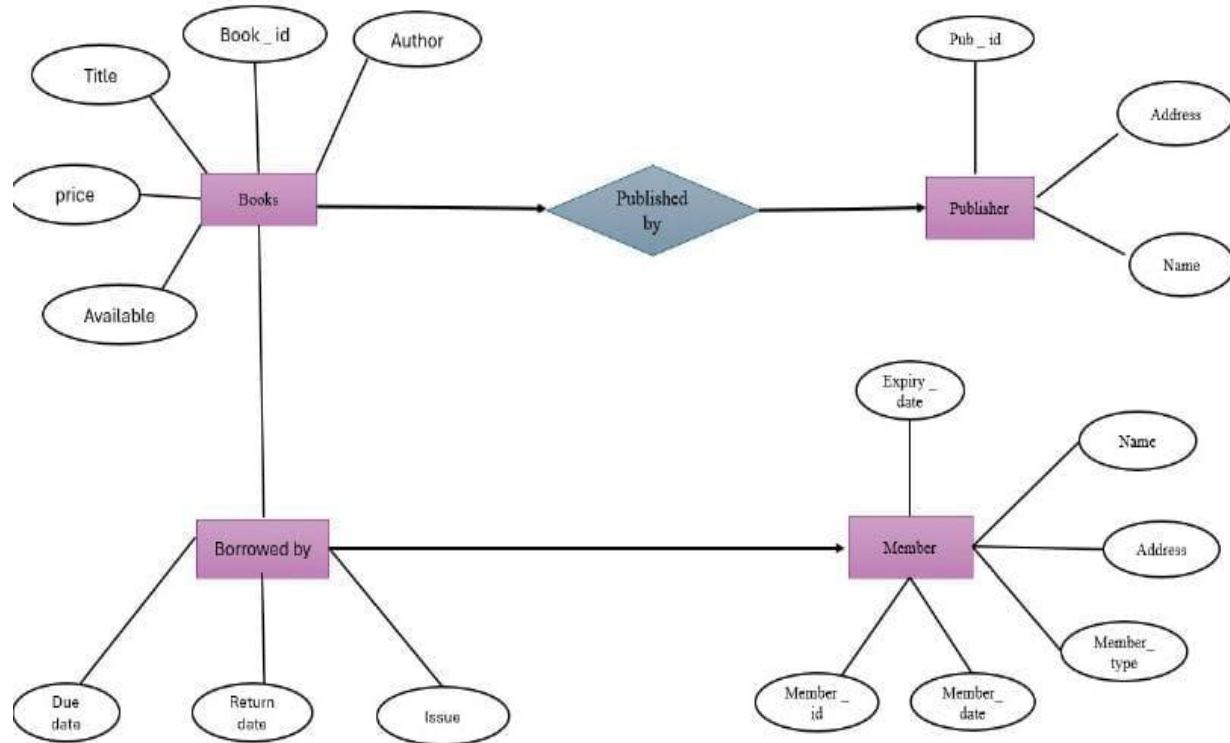
Architecture Diagram Explanation

1. Library Management System (Main System): Acts as the core system that manages and connects all other modules.
2. Book_id, which uniquely identifies each book.
3. Book _ id: A unique identifier that links books to members and transactions.
4. Members : Stores details about users registered in the library.
5. Transaction : Captures book borrow/return activities . It is linked to Members and Book _ id to record who borrowed which book.
6. Issue Date & Return Date : Part of the transaction module, representing when a book was issued and when it's due to be returned.
7. Browsing History : Logs user interactions or searches for books .

Schema Diagram



ER diagram



Software Description

Frontend: HTML,CSS

Backend: PHP

DBMS: MySQL

Server: Apache

Results & Discussion

The Library Management System using DBMS efficiently stored and managed data related to books, users, and transactions. It streamlined the issuing and returning process through accurate and fast data retrieval using SQL. Role-based access ensured data security and prevented unauthorized operations. Automated features like due date alerts and fine calculations reduced manual effort. Search and reporting functions improved decision-making and inventory tracking. Overall, the DBMS-based system enhanced performance, reliability, and user satisfaction.

The screenshot shows a web application titled "Library Management System". It features three input fields: "Enter Book Title", "Enter Author Name", and "Enter ISBN Number". Below these fields is a green "Add Book" button. Underneath the button is a section titled "Library Books:" which displays a book entry: "Wings of fire" by "Author: Edison" with "ISBN: 2300056". At the bottom of the page, there is a copyright notice: "© 2025 Library Management System".

The screenshot shows a database management tool interface. At the top, it displays the database name "library_db" and the table "books". Below this, a SQL query is shown: `SELECT * FROM 'books'`. The results of the query are displayed in a table with columns "id", "title", and "author". The table contains two rows: "9 wings of fire" by "APJ" and "10 way of success" by "abc". The interface includes various controls such as "Show all", "Number of rows", "Filter rows", and "Sort by key". At the bottom, there is a section for "Query results operations" with options like "Print", "Copy to clipboard", "Export", "Display chart", and "Create view".

Conclusion

In conclusion, the Library Management System (LMS) is a vital tool for modernizing and automating the operations of a library. The transition from traditional manual methods to a DBMS-powered LMS not only enhances the overall functionality of libraries but also significantly improves the efficiency, accuracy, and security of managing library resources. By centralizing all information related to books, members, transactions, and reports into a digital database, the LMS eliminates the errors, delays, and difficulties inherent in paper-based systems. The use of a relational database ensures data consistency and integrity, allowing libraries to maintain a well-organized catalog that is easily accessible and searchable. The system's modular design, with distinct components such as book management, member registration, book issuance and return tracking, fine management, and reporting, allows for a comprehensive approach to library operations.

Reference

- [1] A. Al-Fuqaha, M. Guizani, M. Mohammadi, M. Aledhari, and M. Ayyash, “Internet of Things: A survey on enabling technologies, protocols, and applications,” *IEEE Communications Surveys & Tutorials*, vol. 17, no. 4, pp. 2347–2376, 2015, doi: 10.1109/COMST.2015.2444095.

- [2] M. A. Jabbar, B. B. Meshram, and S. M. Kharat, “Digital library and its accessibility via cloud computing,” in *Proc. IEEE Int. Conf. on Communication Systems and Network Technologies (CSNT)*, Rajkot, India, 2013, pp. 518–522, doi: 10.1109/CSNT.2013.111.

- [3] R. Manoharan and R. Dhanasekaran, “Implementation of RFID technology in library management system,” *International Journal of Computer Applications*, vol. 69, no. 9, pp. 21–25, May 2013, doi:10.5120/11852-7419.